

DAX MEASURES

Kenya Mobile Money & Agent Network Analytics

This document lists all DAX measures created for the Power BI dashboard, grouped by purpose, with descriptions and formatting notes. Measures are listed in dependency order; several higher-level measures (e.g., Net Revenue) reference base measures (e.g., Total Revenue), so base measures should be created first.

All measures are created on the **FactTransactions** table unless otherwise noted.

1. Core Financial Measures

Total Revenue

Total Revenue = SUM(FactTransactions[Commission])

Description: Sums all commission earned across transactions. This is the platform's gross revenue.

Total Processing Cost

Total Processing Cost = SUM(FactTransactions[ProcessingCost])

Description: Sums the operational cost of processing each transaction.

Net Revenue

Net Revenue = [Total Revenue] - [Total Processing Cost]

Description: Revenue remaining after processing costs are deducted — the platform's true profit contribution. This is the single most important measure in the model; most other insights reference it directly or indirectly.

Total Transaction Value

Total Transaction Value = SUM(FactTransactions[TransactionAmount])

Description: Sums the full monetary value moved through the platform (not revenue, this is the underlying transaction amounts customers sent/withdrew/deposited).

2. Volume & Average Measures

Transaction Count

Transaction Count = COUNTROWS(FactTransactions)

Description: Total number of transactions recorded, regardless of status (success/failed/reversed).

Average Transaction Value

Average Transaction Value = DIVIDE([Total Transaction Value], [Transaction Count])

Description: Mean transaction size across the network. Uses DIVIDE() rather than / to safely return blank instead of an error if Transaction Count is ever zero (e.g., under an extreme filter with no data).

3. Rate & Margin Measures

Success Rate

Success Rate =

*DIVIDE(
CALCULATE(COUNTROWS(FactTransactions), FactTransactions[TransactionStatus]="Success"),
COUNTROWS(FactTransactions)
)*

Description: Percentage of transactions that completed successfully. CALCULATE overrides the row context to count only rows where status = "Success", then divides by all transactions. **Format:** Percentage, 2 decimal places **Note:** Do not multiply by 100 in the formula, apply Percentage number formatting instead, or the value will display incorrectly (e.g., "9598%" instead of "95.98%").

Net Revenue Margin

Net Revenue Margin = DIVIDE([Net Revenue], [Total Revenue])

Description: Percentage of gross revenue retained as net profit after costs. This is the headline profitability KPI for the Executive Overview page. **Format:** Percentage, 2 decimal places

Revenue per Active Agent

Revenue per Active Agent =

*DIVIDE(
[Net Revenue],
CALCULATE(DISTINCTCOUNT(FactTransactions[AgentID]),
FactTransactions[TransactionStatus]="Success")
)*

Description: The KPI that most directly answers the project's central business question, does network growth translate into per-agent profitability, or does adding agents dilute individual earnings? DISTINCTCOUNT ensures each agent is counted once regardless of how many transactions they

processed, and the filter restricts the count to agents with at least one successful transaction ("active" agents).

4. Status Breakdown Measures

(Used primarily to power the Success Rate tooltip page)

Successful Transactions

Successful Transactions =

`CALCULATE(COUNTROWS(FactTransactions), FactTransactions[TransactionStatus]="Success")`

Description: Raw count of successful transactions.

Failed Transactions

Failed Transactions =

`CALCULATE(COUNTROWS(FactTransactions), FactTransactions[TransactionStatus]="Failed")`

Description: Raw count of failed transactions.

Reversed Transactions

Reversed Transactions =

`CALCULATE(COUNTROWS(FactTransactions), FactTransactions[TransactionStatus]="Reversed")`

Description: Raw count of reversed transactions. **Format:** Whole Number

5. Geographic Callout Measures

(Used in the "Top County" cards on the Geographic Intelligence page)

Top County Revenue

Top County Revenue =

`VAR TopCounty =`

`TOPN(1, ALL(DimLocation[County]), CALCULATE([Net Revenue]), DESC)`

`RETURN`

`MAXX(TopCounty, DimLocation[County])`

Description: Returns the *name* of the county with the highest net revenue. `ALL(DimLocation[County])` removes any existing filter on County so the ranking always considers every county, even if a slicer is applied elsewhere. `TOPN(1, ..., DESC)` finds the single highest-ranked row; `MAXX` extracts the county name from that one-row table.

Top County Value

Top County Value =

```
CALCULATE(
    [Net Revenue],
    TOPN(1, ALL(DimLocation[County]), CALCULATE([Net Revenue]), DESC)
)
```

Description: Returns the actual net revenue *value* for the top-ranked county (companion measure to Top County Revenue above, displayed as two side-by-side cards: county name + county value).

6. Dynamic Insight Measure

(Used in the text callout card on the Executive Overview page)

Dynamic Insight

Dynamic Insight =

"M-Pesa leads with " & FORMAT([Net Revenue Margin], "0%") & " margin overall."

Description: Builds a plain-language sentence that updates automatically if the underlying Net Revenue Margin changes (e.g., due to filtering). Concatenates static text with a formatted measure value using &. FORMAT() converts the numeric percentage into a display string so it can be joined with text. **Format:** Text

Note: This version hardcodes "M-Pesa" as the leading provider. A more advanced version could use TOPN/VAR (similar to the Top County measures above) to dynamically identify whichever provider is actually leading under the current filter context, rather than assuming it's always M-Pesa.

Summary Table

Measure	Category	Depends On
Total Revenue	Core Financial	—
Total Processing Cost	Core Financial	—
Net Revenue	Core Financial	Total Revenue, Total Processing Cost
Total Transaction Value	Core Financial	—
Transaction Count	Volume	—
Average Transaction Value	Volume	Total Transaction Value, Transaction Count
Success Rate	Rate/Margin	—
Net Revenue Margin	Rate/Margin	Net Revenue, Total Revenue

Revenue per Active Agent	Rate/Margin	Net Revenue
Successful Transactions	Status Breakdown	—
Failed Transactions	Status Breakdown	—
Reversed Transactions	Status Breakdown	—
Top County Revenue	Geographic Callout	Net Revenue
Top County Value	Geographic Callout	Net Revenue
Dynamic Insight	Insight Text	Net Revenue Margin

Total: 15 measures